Multimodal AI Teaching Copilot: Live Slide Generation from Lecture Speech

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***Abstract*—Prepared slides cannot show what a teacher adds while explaining, and a large language model (LLM) that regenerates slides from a growing transcript rewrites what students are reading. This paper presents a teaching copilot that builds a structured, continuity-aware slide deck on the classroom projector from the teacher's live speech. Locally transcribed speech is sent in bounded units to an LLM that only returns a schema-validated interpretation; a deterministic planner turns it into slide operations that only extend the deck, so the LLM cannot rewrite what students have read, and a transcript-coverage guard restores facts the LLM omits. Across 36 held-out lectures (409 min), run on one laptop with 4 GB of graphics memory, the median delay from speech to slide was 11.2 s. On an annotated subset of 18 lectures, slide changes matched topic boundaries with precision 0.78, 86.6 % of key facts reached a slide, 0.9 % of statements were unsupported by the speech, and five teachers gave mean ratings of 3.87–4.17 of 5 on four criteria. On the same units, an LLM-controlled baseline rewrote or removed 38.2 % of the statements students had already seen, against 2.0 % (lower in all 18 lectures, $p < 0.001$), and a replay ablation showed that opening a slide per concept lowers precision to 0.37.**

***Index Terms*—Automatic slide generation, educational technology, large language models, live systems, speech recognition, human-in-the-loop**

I. Introduction

Slides prepared before a lecture fix its order in advance, so the examples and answers a teacher adds while explaining never reach the screen. Narration paired with relevant pictures helps learners more than narration by itself [1], provided the visuals add no extraneous load [2], [3]; yet no teacher can draw such visuals while teaching.

Existing tools solve only parts of this problem: live captions show verbatim text without the structure of the lesson [4]; slide generators work offline from a finished document or prompt [5]–[11]; and a large language model (LLM) [12], [13] applied to a growing transcript rewrites slides while students read them and adds untaught content [14]. That leaves three gaps. What is projected does not follow what is being taught (P1). Live speech is never turned into a structured visual lesson (P2). And LLM output is neither stable nor faithful to the teacher's words (P3).

The Multimodal AI Teaching Copilot was built to close these gaps. It listens to the teacher and, while the teacher is still talking, puts typed layouts, typeset formulas, diagrams and retrieved images on the projector. The contributions are: (1) incremental slide generation from live speech; (2) an extend-only update policy: the LLM only labels bounded, schema-validated units, and a deterministic planner executes only operations that add to the deck, so the LLM cannot rewrite what students have read; (3) a transcript-coverage guard that restores facts the LLM omits, and a formula guard; and (4) a held-out evaluation with annotation, teacher ratings, a replay ablation and a paired comparison with an LLM-controlled baseline.

II. Related Work

A. Multimedia Learning and Lecture Speech

In Mayer's experiments, learners given text with matching pictures outperformed those given text only [1]; Sweller attributes such effects to how little a learner can hold in mind at once [2], [3], and decorative pictures reduced test scores in PowerPoint lectures [15]. The system therefore uses short typed layouts and adds an image only for a concrete object that is taught, vetted with CLIP, which Radford *et al.* trained on image–caption pairs so that any picture can be scored against a text prompt [16].

Whisper by Radford *et al.* [17], run with 8-bit weights by faster-whisper [18], and Silero VAD [19] form the speech front end. Kushalnagar *et al.* showed that verbatim real-time captions are hard for students to follow [4], and Gonzalez *et al.* found that LLM summaries of recorded lectures improved quiz scores [20]; both produce text, not a structured visual lesson (P2).

B. Segmentation, Slide Generation and LLMs

TextTiling by Hearst places subtopic boundaries where the similarity of adjacent blocks drops [21]; Sentence-BERT by Reimers and Gurevych compares sentence embeddings by cosine similarity [22], and Wang *et al.* distilled it into MiniLM [23]. These methods segment a complete transcript offline; here the similarity drop is one online signal. Boundaries are scored with the P_k measure of Beeferman *et al.* [24], and fact coverage with the word-recall overlap that ROUGE-1 by Lin uses for summaries [25].

PPSGen picks the sentences of a paper that go on slides with integer linear programming [7]. D2S writes text for each slide title from a document [6], and DOC2PPT adds figures as well [5]. LLM systems go further: Xie *et al.* ground educational slides in retrieved sources [8], SlideBot splits the planning across agents [9], PPTAgent edits reference slides [10], and AutoPresent writes slides as code [11]. All of them build the whole deck at once from clean written input; a lecture arrives sentence by sentence as noisy speech (P1, P3).

TABLE I
COMPARISON OF THE PROPOSED SYSTEM WITH PRIOR SYSTEMS

System	Input	Produced	Topic structure	Faithfulness to source	Slide sequence
Live captions [4], [17]	Speech	Live	None (text stream)	Verbatim text	None
Lecture summaries [20]	Recorded lecture	Offline	None (summary)	LLM summary	None
PPSGen, D2S, DOC2PPT [5]–[7]	Paper / document	Offline	Document sections	Extracted from document	Whole deck at once
LLM slide generators [8]–[11]	Topic, document	Offline	LLM outline / plan	Retrieved sources or none	Whole deck at once
Proposed system	**Live speech**	**During lecture**	**Online topic / subtopic**	**Coverage + formula guards**	**Extend-only + teacher**

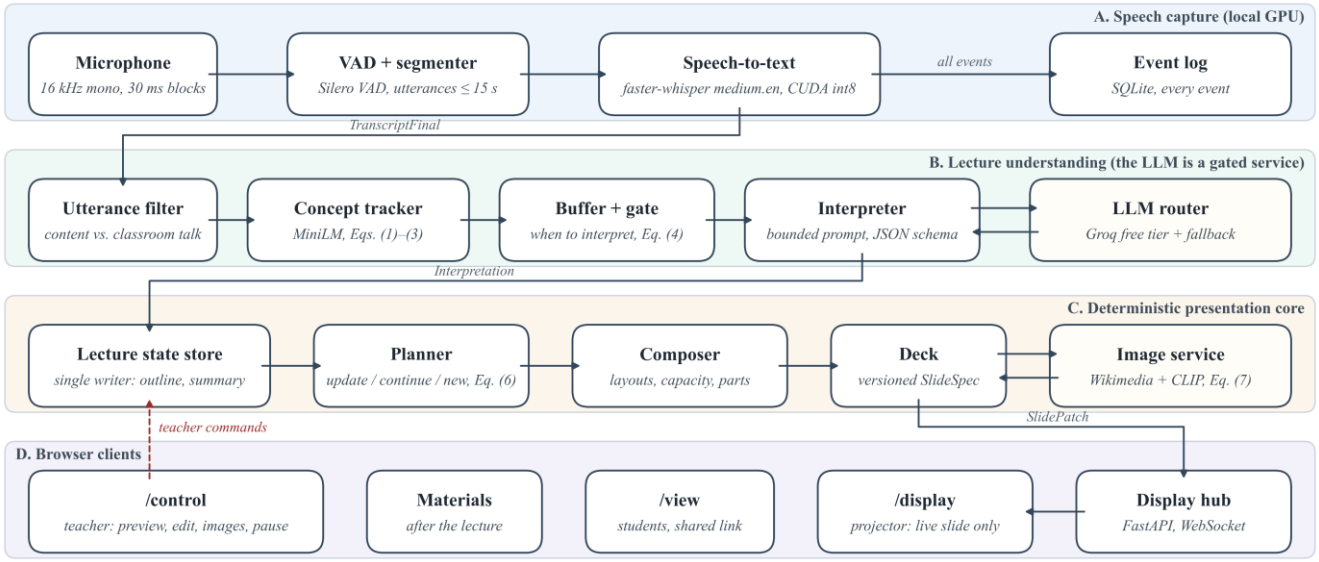


Fig. 1. Implemented architecture. Lanes A–C share one Python process; only the interpreter talks to the LLM.

Kasneci *et al.* argue that teachers must stay in charge when LLMs are used in class [12]; here the teacher can override every automatic decision. Ji *et al.* surveyed hallucination [14], and Willard and Louf constrained generation to a JSON schema [26], which fixes the format but not the facts, so the output is also checked against the transcript. Open-weight models, gpt-oss-120b [13] and the Qwen3 family [27], serve as interpreters on a free hosted tier; generated quizzes follow the review of Kurdi *et al.* [28]. Annotators are compared with Cohen's κ [29] and Gwet's AC1, which holds up when one label dominates [30]; teacher ratings with the ICC(2,1) of Shrout and Fleiss [31].

Table I puts these systems side by side. None of them takes streaming speech and decides, as the lecture goes on, which slide to change while students are reading it. That gap is the subject of this work.

III. METHODOLOGY

A. System Overview

Fig. 1 shows the pipeline. Speech is captured and recognized, filler is dropped, and a concept tracker follows the topic. A gate waits for complete sentences and passes them on as one unit, which the LLM interprets with bounded context. The planner and composer, both deterministic, turn the answer into slide operations and layouts, and the result goes to the projector, the teacher and the students. Four rules were fixed early: the LLM supplies labels but never executes a slide operation; it never sees the whole transcript; a new concept is not by itself a new slide; and user interfaces send commands rather than change the state.

All components run in one Python *asyncio* process and exchange typed events; every event is logged in SQLite, from which each lecture is replayed for evaluation and materials. Recognition runs on the GPU, MiniLM and CLIP on the CPU (ONNX), and the LLM on the Groq free tier (openai/gpt-oss-120b [13]; qwen/qwen3.8-27b of the Qwen3 family [27] when that quota is spent; Oct. 2026), as a 7B-class local model does not fit beside the speech model in 4 GB. Of the 1,619 held-out units, 71.6 % were interpreted by the first model, 24.0 % by the second and 4.4 % by the deterministic fallback.

B. Speech Capture and Concept Tracking

Silero VAD [19] segments 16 kHz audio (utterances up to 15 s), and faster-whisper [18] with Whisper medium.en [17] (int8, beam size 1, keyword prompt) transcribes them. A deterministic filter removes classroom management, meta talk and fillers. Each content utterance is embedded with MiniLM [23]; with the L2-normalized embedding e_t and concept centroid c_{t-1}, the shift score is the cosine distance $s_t = 1 - e_t \cdot c_{t-1}$ used with Sentence-BERT [22], and the centroid is the exponentially weighted mean $c_t = \mathrm{norm}((1-\alpha)c_{t-1} + \alpha e_t)$ with $\alpha = 0.35$, raised to 0.6 at a boundary so it re-seeds on the new concept. Following the similarity-drop criterion of TextTiling [21], a boundary is signalled when

$$b_t = \kappa_t \vee (s_t \geq \tau), \tau = 0.75 \quad (1)$$

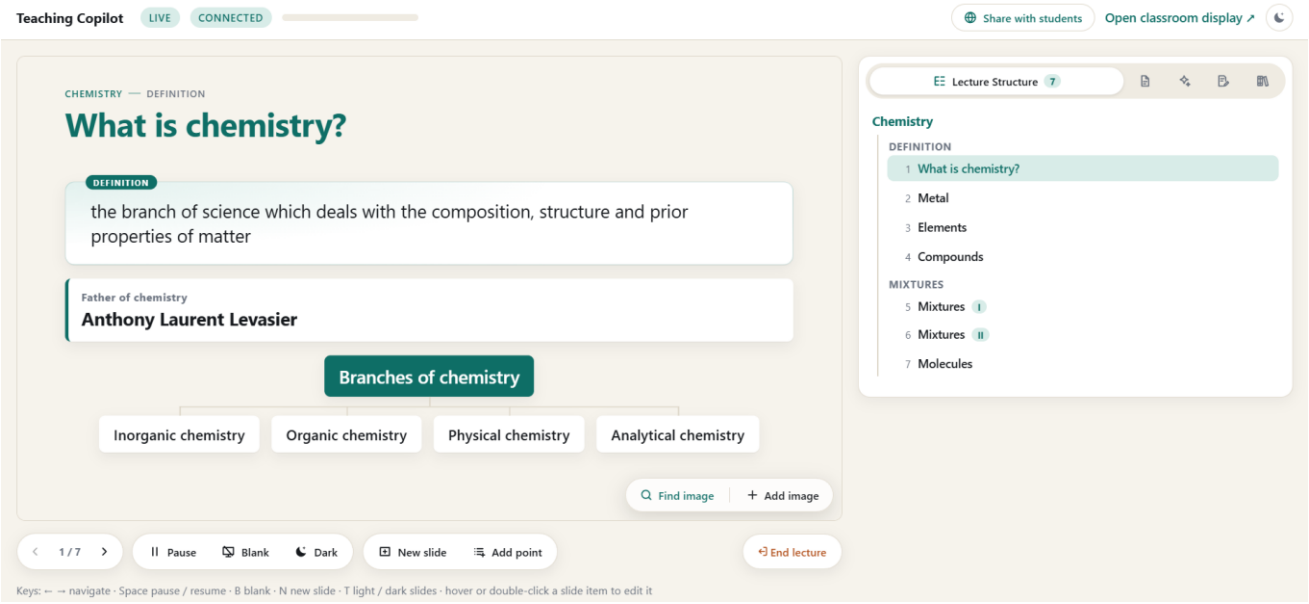


Fig. 2. Teacher control view (/control) in a live chemistry lecture: the projected slide as an editable preview with image tools and lecture controls (left) and the lecture structure tree built by the system, topic → facet → slides, with overflow parts I and II (right).

where κ_t marks a cue phrase ("moving on to", "next"); τ is high because shift scores within one facet were 0.3–0.6.

C. Gate and Token Budget

Content accumulates in a buffer until a rule-based gate fires. With p_t the silence after a finished sentence, w_t the buffered words and a_t the age of the oldest pending line (s),

$$G_t = (p_t \geq 1.2 \wedge w_t \geq 6) \vee w_t \geq 40 \vee b_t \vee a_t \geq 12 \quad (2)$$

with an 8 s floor between calls, set by the free quota (200,000 tokens per key and model per day).

D. Bounded Interpretation and Transcript-Grounding Safeguards

Each prompt carries the subject and level, an outline of at most 150 tokens, the current slide with its items numbered, a rolling summary (at most 120 tokens) and the new lines, under 3,000 tokens in all (temperature 0.2). The answer is JSON: topic and subtopic, how the unit relates to what came before (same concept, elaboration, sub-concept, sibling, new topic or digression), typed content, an image hint, and *concerns*, i.e. statements the model suspects are wrong. Pydantic checks it; if one repair attempt also fails, a deterministic fallback just shows the spoken sentences as points. Because the model sometimes drops a fact, a coverage guard compares every content sentence u of six or more words with the output O. With $W(\cdot)$ the set of word stems and a recall-style overlap as in ROUGE-1 [25],

$$|W(u) \cap W(O)| \leq \tfrac{1}{3}|W(u)| \Rightarrow u \text{ is shown as said} \quad (3)$$

a sentence the model left out is put back on the slide. Formulas have their own guard, which flags any changed token. A correction is accepted only above a confidence of 0.75 for a factual fix or 0.4 for a mis-hearing; below that the slide keeps what was said, and the teacher gets a card with both versions. None of this proves a statement correct; it only keeps the slide faithful to the transcript.

E. Slide Planner

For each interpretation the planner picks one operation: *update* the working slide, *continue* on a new slide of the same topic, open a *new* topic slide, *retitle*, or do nothing. No operation lets the LLM rewrite or delete something already shown. Shown text changes in three cases only: the composer merges an elaboration into the item it elaborates, a mis-hearing is corrected, or the teacher edits it. One unit can be misread, so a topic change has to be confirmed:

$$C_n = B_n \vee \left(\bar{s}_n^{\max} \geq \tau\right) \vee \left(T_n \equiv \tilde{T}_{n-1}\right) \quad (4)$$

where B_n is true if a boundary (1) occurred in unit n, $\bar{s}_n^{\max}$ is its largest shift score smoothed over two utterances, T_n its topic and $\tilde{T}_{n-1}$ an unconfirmed candidate from the previous unit. Unconfirmed content is shown tentatively and moved once confirmed. A minimum dwell of 15 s (6 s for the next part of a full slide) prevents rapid changes, near-duplicates (similarity ≥ 0.85) are merged, and teacher edits are never overwritten. All thresholds were set by hand on D1 and are tested by ablation in Section IV.

F. Composer and Image Retrieval

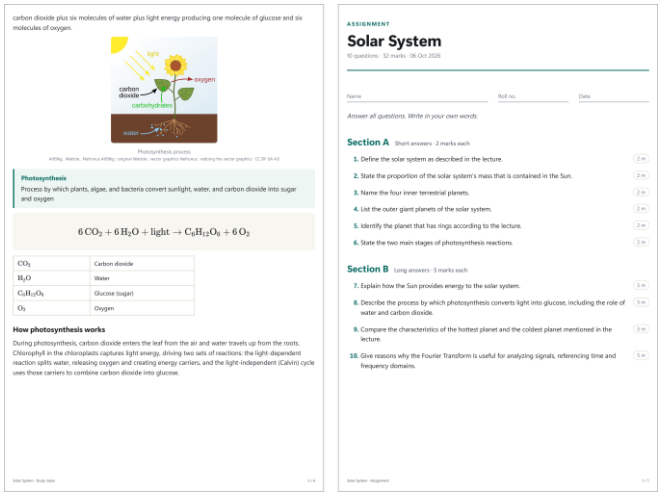
Items become definition cards, key points, fact tiles, process flows, comparison tables, timelines, cause-effect chains, classification trees or formulas. A full slide continues in a further part (I, II, …); text is never shrunk to fit, and spoken mathematics is typeset with KaTeX. For a concrete object the LLM suggests a query; an openly licensed Wikimedia image (source, author and licence kept) is placed only if CLIP's zero-shot similarity [16] ranks it above generic distractor prompts ("a logo", "a map") and the query matches the slide text.

G. Live Display and Teacher Control

A FastAPI server pushes versioned slide patches over WebSocket to /display (projector), /control (teacher) and /view (students, shared link). The control view (Fig. 2) adds editing, image search and upload, pause, blank screen, mistake cards, the running transcript and the lecture structure tree.

H. Lecture Materials

After the lecture the event log is reused: on request it becomes study notes, an assignment (Fig. 3), a PPTX deck, or a live quiz or test answered through the shared link, all from the taught content and edited by the teacher. These features are not evaluated here.



(a) Study notes (b) Assignment

Fig. 3. Materials generated from the event log of a recorded D1 lecture (not evaluated): (a) a study-notes page; (b) the assignment.

IV. Experimental Evaluation

A. Setup and Data

Every run used one laptop (NVIDIA RTX 3050, 4 GB video memory; 16 GB system memory). D1 is six lectures spoken by the authors (114.1 min). The held-out set D2 (Tables II and III) has 30 lectures spoken live, unscripted, by six speakers into the laptop microphone (Recorded, physics to economics) and six 10-min public lectures played in as audio in real time (Public): MIT OpenCourseWare 7.016, 5.111, 18.01 and 8.01 (CC BY-NC-SA) and NPTEL Computer Networks and History of Economic Theory (CC BY-SA). The speakers consented to the recording. Against MIT's human captions of the 7.016 lecture (1,257 words), the live transcript has a word error rate of 7.1 %. Teachers did not edit slide text during a test; in D2 they deleted two items and accepted corrections from concern cards in two lectures. Three populations are evaluated: all 36 lectures for latency and operation, an annotated subset of 18 (12 Recorded, 6 Public; 191 min) for accuracy and the ablation, and 30 sampled slides for teacher ratings.

TABLE II
Evaluation Data

Set	Lectures	Minutes	Units	Fallback units	Slides
D1 (development)	6	114.1	389	45	163
D2 Recorded	30	349.2	1,275	59	453
D2 Public	6	60.0	344	13	52
D2 all (held out)	**36**	**409.2**	**1,619**	**72**	**505**

Units: groups of finished sentences sent for interpretation; fallback units were interpreted without the LLM (quota spent or provider failure).

TABLE III
Live System Performance (Session Logs)

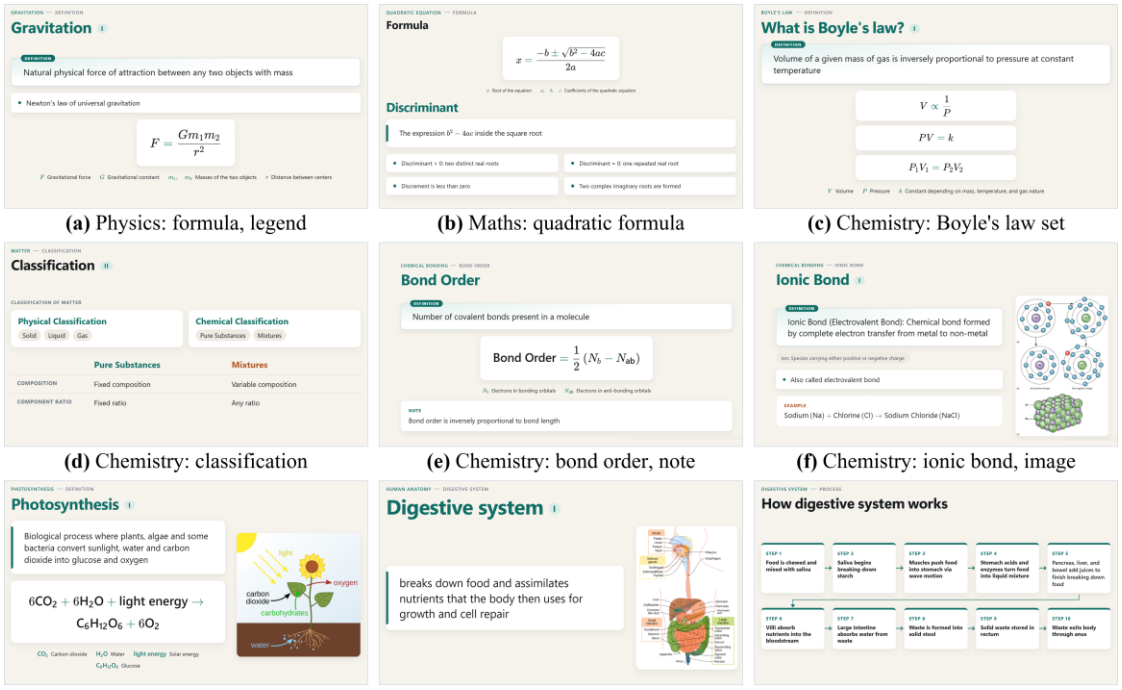
Measure (s)	D1	D2 Rec.	D2 Pub.	D2 all
Speech-to-text per utterance	0.66	0.71	0.60	**0.68**
LLM call	1.35	1.92	2.23	**1.96**
Speech to slide, median	12.5	10.1	17.1	**11.2**
Speech to slide, 95th percentile	36.8	23.2	93.4	**53.0**

Medians unless stated; speech to slide: from the end of a spoken line to its display. Public includes web4 and web6 (provider outages).

Annotation. Two authors (A and B) independently annotated the 18 lectures, chosen to mix subjects and speakers with 6–17 min each. Before seeing the slides, each marked topic boundaries (where a teacher would start a new board) and key facts in the transcript, then judged whether each key fact was on a slide and whether each slide statement was said and correct; scores are the mean of A and B, without adjudication. A system boundary is the utterance that opened a new content slide. Boundaries are scored by precision, recall and F1 within ±1 utterance, averaged per lecture, and by P_k [24], the probability that two utterances half a segment apart are wrongly split or joined; agreement is Cohen's κ [29] and, for skewed labels, Gwet's AC1 [30]. Intervals are 95 % bootstrap intervals over lectures. *Ratings.* Five teachers (T1–T5), not involved in development, rated 30 random D2 slides, each shown with what was said, from 1 to 5 on four criteria; their consistency is the ICC(2,1) of Shrout and Fleiss [31].

B. Topic Boundaries and Content Accuracy

Slides change where the topic changes (Table IV): boundary precision is 0.78 (95 % interval 0.68–0.88), so a projected slide is rarely replaced within a topic, and P_k is 0.32. Recall is 0.42 (0.30–0.55): a new concept is not a new slide, so the planner merges into the live slide sub-points that annotators marked as boundaries, most often in the Public lectures (recall 0.19); Section IV-C shows this is a trade-off. Of 669 key facts, 86.6 % (84.6–88.5 %) reached a slide (partly shown counts ½), and only 0.9 % (0.2–1.8 %) of 2,504 statement judgments were not supported by the speech. The coverage guard (3) is needed for this: 60 of the covered facts reached a slide only through a sentence it restored, so without it coverage would be 77.7 %; the guard adds 9.0 points (6.4–11.7). Of the 3.7 % marked wrong, 41 of 79 were also marked as said (a slip or mis-hearing): slides follow the speech, so transcript support is not factual correctness. Agreement is good for boundaries (κ = 0.72) and high for "supported" (88.8 %, AC1 0.88); κ is only 0.07 there because nearly all statements were supported, which lowers κ but not AC1 [30].



(a) Physics: formula, legend **(b)** Maths: quadratic formula **(c)** Chemistry: Boyle's law set

(d) Chemistry: classification **(e)** Chemistry: bond order, note **(f)** Chemistry: ionic bond, image

(g) Biology: reaction, image **(h)** Biology: labelled image **(i)** Biology: ten-step process

Fig. 4. Slides from development lectures by the authors; (a), (c), (i) re-render recorded LLM answers, the others are app screenshots. Text is as spoken (the slip "Discrement" in (b)).

Test29 ran entirely on the fallback (quota spent). It still covered 83.1 % of key facts with 0.7 % unsupported statements, but it rarely opened a slide (recall 0.04) and 11 of its 13 slides were plain points, so the fallback keeps the content and the LLM adds the structure. The 17 LLM lectures reach precision 0.77 and recall 0.45. Teachers rated the slides 3.87–4.17 of 5 (ICC(2,1) 0.45–0.56, moderate). In Fig. 4 the layout follows the kind of content, not the subject.

TABLE IV
ACCURACY AND TEACHER RATINGS (18 ANNOTATED D2 LECTURES)

Measure	Recorded	Public	All
Topic boundaries			
Precision ↑	0.81	0.73	**0.78**
Recall ↑	0.54	0.19	**0.42**
F1 ↑	0.59	0.29	**0.49**
P_k ↓	0.29	0.37	**0.32**
Slide content			
Key facts on a slide ↑	88.0 %	84.0 %	**86.6 %**
without coverage guard	81.9 %	69.1 %	77.7 %
Statements not said ↓	0.9 %	0.9 %	**0.9 %**
Statements marked wrong ↓	3.6 %	4.0 %	**3.7 %**
Teacher ratings (1–5) ↑			
Correctness	4.20	4.06	**4.15**
Relevance	3.94	3.80	**3.89**
Readability	4.12	4.26	**4.17**
Usefulness	3.93	3.76	**3.87**
Annotator agreement			
Boundaries, κ (raw)	0.81 (94 %)	0.56 (90 %)	0.72 (92 %)
Statement said, AC1 (raw)	0.89 (90 %)	0.84 (85 %)	0.88 (89 %)

↑ / ↓: higher / lower is better. Mean of annotators A and B; boundaries within ±1 utterance; ratings: T1–T5 mean over 30 slides.

C. Ablation of the Planner Rules

Table V replays the recorded signals and LLM answers of the 18 lectures through the planner, changing one rule at a time; no LLM is called, and the unchanged replay reproduces the live decks. Most of the effect comes from the continuity rule: if every concept opens its own slide, the deck changes 3.2 times a minute, 200 slides are replaced within 10 s of appearing, and precision falls to 0.37 (P_k 0.47). The dwell time only delays a change, so without it changes per minute stay at 1.15, yet slides replaced within 10 s rise from 16 to 25. A dwell of up to 20 s costs nothing in precision or recall; at 45 s they drop to 0.67 and 0.32. Topic confirmation made no measurable difference on D2 (precision 0.79 without it) and stays only as a safeguard. When the LLM's labels decide directly but slides are still only extended, boundaries hardly change, so the stability comes from the extend-only operations, not from the labels.

TABLE V
PLANNER ABLATION BY REPLAY (18 ANNOTATED D2 LECTURES)

Planner variant	Precision ↑	Recall ↑	P_k ↓	Changes per min	Replaced < 10 s ↓
Full planner (= live)	**0.78**	0.42	**0.32**	**1.15**	16
No dwell time	0.78	0.42	0.32	1.15	25
No topic confirmation	0.79	0.45	0.31	1.12	10
LLM labels decide	0.78	0.46	0.30	1.17	21
Each concept a slide	0.37	0.86	0.47	3.18	200

Boundary scores as in Table IV; replaced < 10 s: slides replaced within 10 s of being shown.

D. Comparison With an LLM-Controlled Baseline

TABLE VI
PROPOSED SYSTEM VS. LLM-CONTROLLED BASELINE (SAME D2 UNITS)

Measure	Proposed	LLM baseline
Seen statements rewritten/removed ↓	**2.0 %**	38.2 %
Seen statements lost from final deck ↓	**6.2 %**	65.8 %
Lectures with fewer rewrites ↑	**18 of 18**	0 of 18
Boundary precision ↑	**0.78**	0.72
P_k ↓	**0.32**	0.36
Content slides per minute	1.25	0.79

↑ / ↓: higher / lower is better. Rewritten: statements of a replaced slide version that the next version changed or removed; lost: shown statements absent from the final deck.

An LLM-controlled baseline without extend-only operations receives, for the same 850 units of the 18 annotated lectures, the slide it showed last and the last 60 s of transcript, and returns the slide to show, without planner, guards or composer (same two models, 0.5 M tokens). It placed slide changes about as well (Table VI; precision 0.72 vs. 0.78, $p = 0.34$) but rewrote or removed 1,439 of the 3,763 statements students had already seen (38.2 %), against 91 of 4,515 (2.0 %), and 65.8 % of what it showed was gone from its final deck, against 6.2 %. The difference holds in all 18 lectures: the mean per-lecture gap in rewrites is 39 points (95 % interval 32–47; exact paired sign-flip permutation test, $p < 0.001$). Of the proposed system's 91 changes, 24 extended a statement, 6 followed a correction the teacher accepted and 61 were composer merges. The baseline also lacks the guards and composer; Table V isolates the planner rules.

E. End-to-End Latency

Over the 36 D2 lectures (Table III), planning and composing take 2 ms; the median from the end of a spoken line to its display is 11.2 s, 7.6 s of it the gate's deliberate wait for a complete idea: the system is live, not hard real-time. In web4 and web6 the free provider failed (timeouts, HTTP 429), units queued, and the 95th percentile rose to 53.0 s; without them the median is 10.2 s and the 95th percentile 23.4 s.

V. DISCUSSION AND CONCLUSION

With the LLM as a bounded interpreter, the copilot builds a continuity-aware deck from live speech, and extend-only operations keep what students read stable (2.0 % vs. 38.2 % rewritten). Limitations: the annotators were authors (though they marked boundaries and facts before seeing slides); recall is moderate by design; an append-only baseline prompt was not tested; the word error rate comes from the one lecture with human captions; the free provider caused outages; and without a classroom study no claim is made about learning. Anonymized annotations, ratings and evaluation scripts are available on request.

REFERENCES

[1] R. E. Mayer, *Multimedia Learning*, 2nd ed. Cambridge, U.K.: Cambridge Univ. Press, 2009.

[2] J. Sweller, "Cognitive load during problem solving: Effects on learning," *Cognitive Sci.*, vol. 12, no. 2, pp. 257–285, 1988.

[3] R. E. Mayer and R. Moreno, "Nine ways to reduce cognitive load in multimedia learning," *Educ. Psychologist*, vol. 38, no. 1, pp. 43–52, 2003.

[4] R. S. Kushalnagar, W. S. Lasecki, and J. P. Bigham, "Accessibility evaluation of classroom captions," *ACM Trans. Accessible Comput.*, vol. 5, no. 3, Art. no. 7, 2014.

[5] T.-J. Fu *et al.*, "DOC2PPT: Automatic presentation slides generation from scientific documents," in *Proc. AAAI Conf. Artif. Intell.*, vol. 36, no. 1, 2022, pp. 634–642.

[6] E. Sun *et al.*, "D2S: Document-to-slide generation via query-based text summarization," in *Proc. NAACL-HLT*, 2021, pp. 1405–1418.

[7] Y. Hu and X. Wan, "PPSGen: Learning-based presentation slides generation for academic papers," *IEEE Trans. Knowl. Data Eng.*, vol. 27, no. 4, pp. 1085–1097, Apr. 2015.

[8] E. Xie *et al.*, "Leveraging grounded large language models to automate educational presentation generation," in *Proc. Large Found. Models Educ. Assessment*, PMLR, vol. 264, 2025, pp. 207–220.

[9] E. Xie *et al.*, "SlideBot: A multi-agent framework for generating informative, reliable, multi-modal presentations," in *Proc. AAAI Conf. Artif. Intell.*, vol. 40, no. 48, 2026, pp. 40907–40915.

[10] H. Zheng *et al.*, "PPTAgent: Generating and evaluating presentations beyond text-to-slides," in *Proc. EMNLP*, 2025, pp. 14402–14418.

[11] J. Ge *et al.*, "AutoPresent: Designing structured visuals from scratch," in *Proc. IEEE/CVF CVPR*, 2025, pp. 2902–2911.

[12] E. Kasneci *et al.*, "ChatGPT for good? On opportunities and challenges of large language models for education," *Learn. Individual Differences*, vol. 103, Art. no. 102274, 2023.

[13] OpenAI, "gpt-oss-120b & gpt-oss-20b model card," arXiv:2508.10925, 2025.

[14] Z. Ji *et al.*, "Survey of hallucination in natural language generation," *ACM Comput. Surv.*, vol. 55, no. 12, Art. no. 248, 2023.

[15] R. A. Bartsch and K. M. Cobern, "Effectiveness of PowerPoint presentations in lectures," *Comput. Educ.*, vol. 41, no. 1, pp. 77–86, 2003.

[16] A. Radford *et al.*, "Learning transferable visual models from natural language supervision," in *Proc. ICML*, 2021, pp. 8748–8763.

[17] A. Radford *et al.*, "Robust speech recognition via large-scale weak supervision," in *Proc. ICML*, 2023, pp. 28492–28518.

[18] SYSTRAN, "faster-whisper: Faster Whisper transcription with CTranslate2," GitHub. [Online]. Available: https://github.com/SYSTRAN/faster-whisper

[19] Silero Team, "Silero VAD: Pre-trained enterprise-grade voice activity detector," GitHub, 2021. [Online]. Available: https://github.com/snakers4/silero-vad

[20] H. Gonzalez *et al.*, "Automatically generated summaries of video lectures may enhance students' learning experience," in *Proc. 18th Workshop BEA*, 2023, pp. 382–393.

[21] M. A. Hearst, "TextTiling: Segmenting text into multi-paragraph subtopic passages," *Comput. Linguistics*, vol. 23, no. 1, pp. 33–64, 1997.

[22] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," in *Proc. EMNLP-IJCNLP*, 2019, pp. 3982–3992.

[23] W. Wang *et al.*, "MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained transformers," in *Proc. NeurIPS*, vol. 33, 2020, pp. 5776–5788.

[24] D. Beeferman, A. Berger, and J. Lafferty, "Statistical models for text segmentation," *Mach. Learn.*, vol. 34, no. 1–3, pp. 177–210, 1999.

[25] C.-Y. Lin, "ROUGE: A package for automatic evaluation of summaries," in *Proc. ACL Workshop Text Summarization*, 2004, pp. 74–81.

[26] B. T. Willard and R. Louf, "Efficient guided generation for large language models," arXiv:2307.09702, 2023.

[27] A. Yang *et al.*, "Qwen3 technical report," arXiv:2505.09388, 2025.

[28] G. Kurdi *et al.*, "A systematic review of automatic question generation for educational purposes," *Int. J. Artif. Intell. Educ.*, vol. 30, no. 1, pp. 121–204, 2020.

[29] J. Cohen, "A coefficient of agreement for nominal scales," *Educ. Psychol. Meas.*, vol. 20, no. 1, pp. 37–46, 1960.

[30] K. L. Gwet, "Computing inter-rater reliability and its variance in the presence of high agreement," *Br. J. Math. Stat. Psychol.*, vol. 61, no. 1, pp. 29–48, 2008.

[31] P. E. Shrout and J. L. Fleiss, "Intraclass correlations: Uses in assessing rater reliability," *Psychol. Bull.*, vol. 86, no. 2, pp. 420–428, 1979.